

CSC621 Data Structures and Algorithm

Lab Exercise 4 - Queues

Question 1

You are tasked with developing a program to manage incoming requests in a web server application. Each request is represented by a unique identifier (ID), and they need to be processed in the order they arrive. Implement a queue data structure using arrays to simulate the following operations:

1. Accept incoming requests and enqueue them into the queue.
2. Process requests by dequeuing them from the front of the queue.
3. Provide functionality to display the current state of the queue, including the IDs of requests waiting to be processed.
4. Implement error handling to prevent dequeuing from an empty queue.

Write a program to simulate the above operations efficiently, considering the following aspects:

- Ensure that the program can handle a large number of incoming requests without compromising performance.
- Display appropriate messages indicating when a request is enqueued, dequeued, or when the queue is empty.
- Consider optimisations to minimize the time complexity of enqueue and dequeue operations.

Your program should be capable of handling various scenarios, such as fluctuating request rates and different queue sizes. Provide comments in your code to explain the logic and functionality of each section.

You may assume the following:

- A unique integer ID represents each request.
- The maximum capacity of the queue is predefined and provided as input.
- The program will be tested with different request arrival rates and queue sizes to assess its efficiency and reliability.
- Write the program code along with any necessary explanations or assumptions made during the implementation.

Question 2

A **palindrome** is a sequence of symbols such as a word, phrase, verse, or sentence, that reads the same way from either direction, forward or backward (e.g. madam). Write a program that reads a **string**, places **each character** of the string in both a **queue** and a **stack**, and then verifies whether the string is a palindrome.